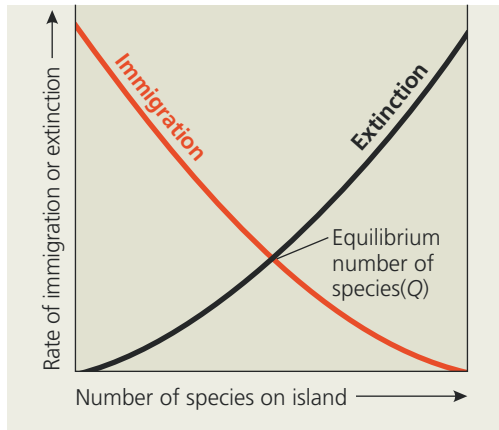


▼ **Figure 54.31 MacArthur and Wilson’s island equilibrium model.** The equilibrium number of species on an island represents a balance between the immigration of new species (red curve) and the extinction of species already there (black curve). The point at which these curves intersect shows the predicted equilibrium number of species (Q).



WHAT IF? Suppose rising sea levels substantially decreased the size of the island. How would that affect (a) the population sizes of species already on the island, (b) the extinction curve shown above, and (c) the predicted equilibrium number of species?

extinction rate *increases*. To see why this is so, consider a newly formed oceanic island that receives colonizing species from a distant mainland. At any given time, an island’s immigration and extinction rates are affected by the number of species currently present. As the number of species already on the island increases, the immigration rate of new species decreases, because any individual reaching the island is less likely to represent a species that is not already present. Moreover, as more species inhabit an island, extinction rates on the island increase because of the greater likelihood of competitive exclusion.

An island’s size and its distance from the mainland are also important. Small islands generally have lower immigration rates because potential colonizers are less likely to reach a small island than a large one. Small islands also have higher extinction rates because they generally contain fewer resources, have less diverse habitats, and have smaller population sizes. With respect to distance, an island that is closer to the mainland generally has a higher immigration rate and a lower extinction rate than one farther away. Arriving colonists increase the abundance of a species on a near island, thereby reducing the species’ chance of extinction.

MacArthur and Wilson’s model is called the *island equilibrium model* because an equilibrium will eventually be reached where the rate of species immigration equals the rate of species extinction. The number of species at this equilibrium point is correlated with the island’s size and distance from the mainland. Although the number of species ultimately should stabilize at a constant level, immigration and extinction continue and hence the exact species composition on the island may change over time.

Researchers tested the island equilibrium model in an experiment on six small mangrove islands in the Florida Keys

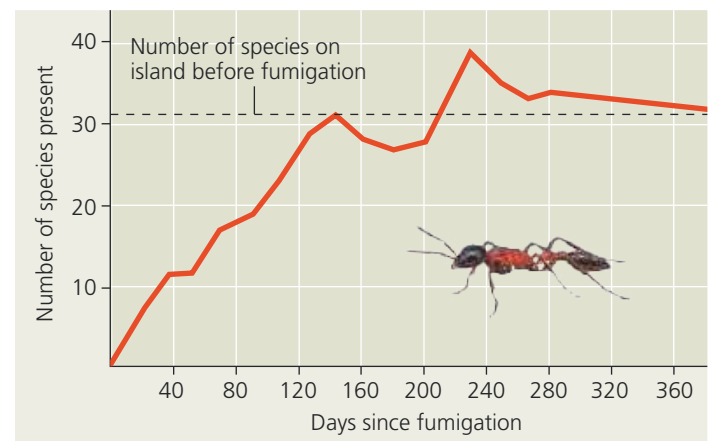
► **Figure 54.32 Mangrove island.** The islands that researchers studied were small, each consisting of one or a few mangrove trees.



(**Figure 54.32**). To do this, they first painstakingly identified and counted all of the arthropod species on each island. As predicted by the model, they found more species on islands that were larger and closer to the mainland. They then fumigated four of the islands with methyl bromide to kill all of the arthropods. As also predicted by the model, the number of arthropod species on the islands increased over time to near their pre-fumigation values (**Figure 54.33**). The island closest to the mainland recovered first, while the island farthest from the mainland was the slowest to recover. The number of arthropod species on the remaining two islands—which were not fumigated and hence served as controls—remained approximately constant during the study.

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▼ **Figure 54.33 Testing the island equilibrium model.** The graph shows results for one of the islands studied. The number of arthropod species increased over time; by 140 days, it had reached levels similar to those found on the island prior to the start of the experiment.



CONCEPT CHECK 54.4

1. Describe two hypotheses as to why species diversity is greater in the tropics than in temperate and polar regions.
2. Describe how an island’s size and distance from the mainland affect the island’s species richness.
3. **WHAT IF?** Based on MacArthur and Wilson’s island equilibrium model, predict how the richness of birds on an island would compare with that of snakes and lizards there.

For suggested answers, see Appendix A.